

GraphSynth: Autonomous Optimization of PyTorch Execution Graphs via Search-Guided Compiler Pass & Triton Kernel Synthesis

Abstract

Static, rule-based lowerings used in MLIR and PyTorch Inductor do not adapt well to new operator topologies, so memory-bound kernels routinely lose the majority of their runtime to redundant DRAM traffic, while writing expert Triton or CUDA kernels by hand takes months of engineering effort per operator. We present GraphSynth, a closed-loop agentic framework that profiles a PyTorch subgraph, synthesizes a candidate Triton kernel for it using an LLM, checks the kernel numerically against the PyTorch reference implementation, and profiles it on real hardware to decide whether to accept the kernel or iterate.

We evaluate GraphSynth on an NVIDIA A100-SXM4-40GB using ten attention variants that PyTorch’s fused-kernel catalog does not cover. The system produced a verified Triton kernel for every one of the ten, running $10.83\times$ faster than eager PyTorch and $3.04\times$ faster than `torch.compile`, measured as geometric means over the ten operators. Two of these kernels are faster than PyTorch’s hand-tuned FlashAttention path, because their masks are sparser than causal attention and a kernel that skips fully masked blocks therefore does less work. PyTorch’s `flex.attention` remains faster on the eight variants it can express, by a geometric mean of $1.81\times$, but it cannot express the remaining two, which replace the softmax normalization instead of modifying the attention scores. GraphSynth therefore offers wider coverage and full automation rather than higher peak performance. As a control, we also ran seven single-tensor operators, where GraphSynth matches `torch.compile` exactly (geometric mean $1.00\times$) and all four paths reach 87 to 88% of peak DRAM bandwidth, showing that the method reports no gain where memory bandwidth already sets the limit.

Index Terms: compiler optimization, arithmetic intensity, Triton, LLM-guided kernel synthesis, GPU roofline analysis.

1. Introduction

Deep learning inference performance is bottlenecked by memory bandwidth utilization rather than raw compute throughput for the majority of operators in modern architectures. Static compiler backends such as PyTorch Inductor and MLIR-based stacks rely on rule-based lowering to a fixed catalog of known operators, and any operator or fusion pattern outside that catalog silently reverts to unfused, sequential execution, incurring redundant DRAM traffic.

Hand-writing an expert Triton or CUDA kernel to close this gap takes an experienced engineer weeks to months per operator, and does not scale as new operator topologies emerge faster than vendor libraries can be hand-tuned. GraphSynth addresses this by closing the loop between LLM-based kernel synthesis and real hardware feedback. An autonomous agent profiles a PyTorch subgraph, synthesizes a Triton kernel candidate from a natural-language operator description, numerically verifies its correctness, and measures it on hardware, iterating until a verified kernel meets the acceptance criterion or a budget is exhausted.

Search-based autotuning frameworks such as Ansor and AutoTVM explore tiling and fusion parameters within a fixed template space, but cannot synthesize new kernel logic. PyTorch’s `flex.attention` occupies an intermediate position: it admits a user-supplied score modification into a hand-tuned FlashAtten-

tion template, which makes it fast but restricts it to patterns expressible as a score transform. An LLM-guided approach instead treats kernel synthesis itself as the search variable.

This paper makes four contributions: (1) a closed-loop PyTorch-to-Triton synthesis workflow with numerical correctness as a mandatory gate and hardware measurement as the optimization signal; (2) an arithmetic-intensity criterion that classifies an operator against the hardware roofline ridge point before any synthesis is attempted; (3) an evaluation against three baselines (eager PyTorch, `torch.compile`, and `flex.attention`), including a single-tensor control experiment that establishes measurement validity; and (4) the observation that a synthesized kernel verified at one shape can be silently incorrect at another.

2. Overview

GraphSynth is structured as four sequential stages executed in a closed loop (Fig. 1): an IR Profiler that classifies operators by arithmetic intensity, an LLM-driven synthesis stage that writes candidate Triton kernels, a numerical verifier that rejects incorrect kernels before they reach performance measurement, and a hardware feedback stage that decides accept-or-iterate. The separation is intentional: profiling limits the search space to kernels where optimization is likely to pay off, verification prevents a fast implementation from being accepted when it changes numerical behavior, and hardware feedback evaluates the generated program on the target GPU rather than a proxy cost model. A candidate is not successful merely because it compiles: it must first match the reference and then satisfy the performance criterion.

3. Methodology

3.1. Platform

All measurements were taken on a single NVIDIA A100-SXM4-40GB (sm80) on Google Compute Engine (a2-highgpu-1g), running PyTorch 2.9.1 with CUDA 12.9 and driver 580.178.04. Peak DRAM bandwidth is $BW_{\text{peak}} = 1555$ GB/s; peak bf16 tensor-core throughput is 312 TFLOP/s and non-tensor-core fp32 is 19.5 TFLOP/s.

3.2. Timing protocol and its validation

An earlier version of this work divided a CUDA-event baseline by an Nsight Compute kernel latency. The two instruments measure different intervals: events around a single call absorb host dispatch gaps, while the Nsight counter reports kernel execution only. A trivial kernel measures $31.7\ \mu\text{s}$ by the former and $\approx 2\ \mu\text{s}$ by the latter, so their ratio inflates any speedup by the dispatch overhead.

All results here use one timing function applied identically to every path: CUDA events around a *loop* of n invocations, with n adaptive so the timed region spans ≈ 300 ms, reporting the median of three repetitions. Bandwidth above 100% of peak is physically impossible and is therefore a sufficient falsifier for a timing method: we calibrate against a device-to-device copy of an 8192×8192 fp32 tensor, which moves exactly 536.9 MB and performs no arithmetic. It measures $389.6\ \mu\text{s}$, i.e. 1376 GB/s or 88.6% of peak. Every bandwidth figure we report is checked against the 100% bound; none exceeds it.

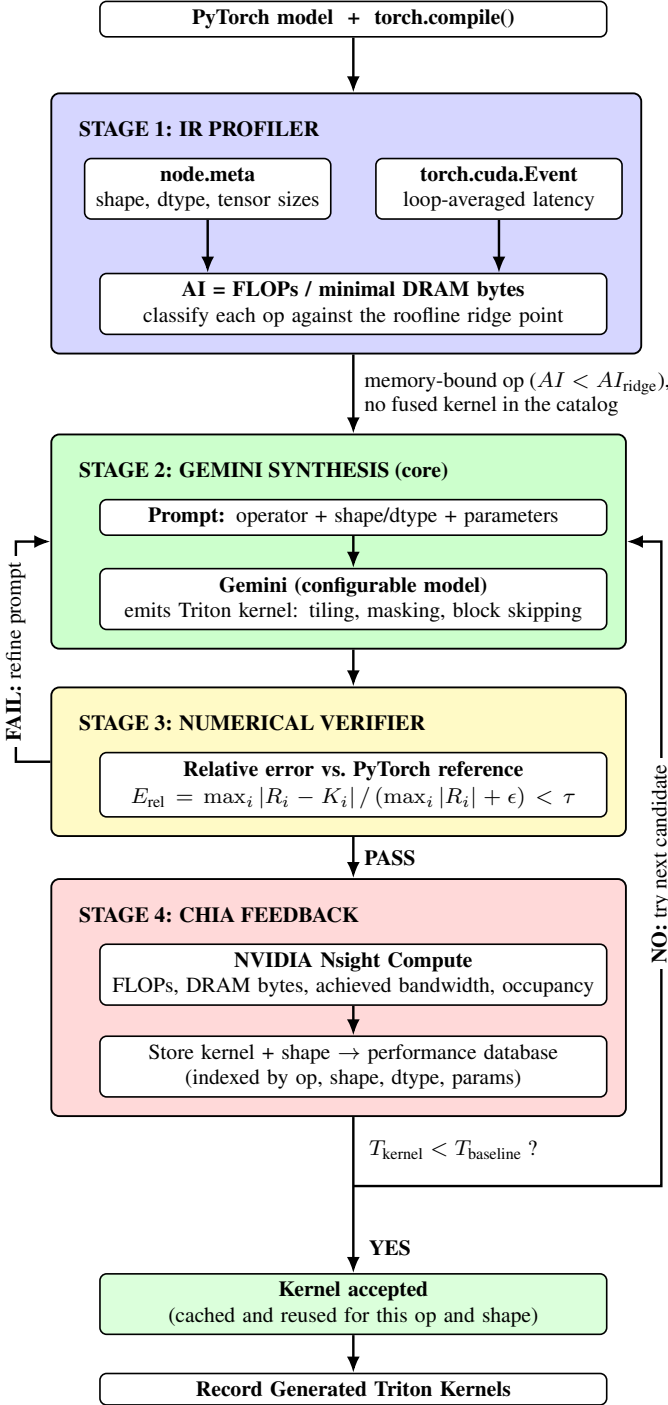


Figure 1: GraphSynth autonomous four-stage synthesis loop. Stage 1 classifies an operator by arithmetic intensity against the device ridge point, Stage 3 gates on relative error against the PyTorch reference, and Stage 4 accepts only a kernel that is both correct and faster than the baseline.

3.3. Correctness gate

For reference output R and kernel output K we require

$$E_{\text{rel}} = \frac{\max_i |R_i - K_i|}{\max_i |R_i| + \epsilon} < \tau, \quad \epsilon = 10^{-6}, \quad (1)$$

with $\tau = 10^{-4}$ for fp32 and 3×10^{-2} for bf16, and no NaN or Inf. A relative criterion is necessary because bf16 attention outputs have magnitude $\approx 10^{-1}$, for which a fixed absolute tolerance would admit kernels with 20% error. Kernels failing the gate are never timed and are reported as failures.

3.4. Arithmetic intensity and Stage 1 classification

An ideal fused kernel reads each input once and writes each output once. We take that minimum traffic $B_{\min}$ as the denominator of bandwidth utilization, $\eta_{\text{BW}} = B_{\min} / (T \cdot \text{BW}_{\text{peak}})$; because $B_{\min}$ assumes perfect reuse, every η_{BW} we report is a lower bound. For a single-tensor operator on an $N \times N$ tensor with b bytes per element, $B_{\min} = 2N^2b$; for attention it counts only the query, key, value and output tensors, never the $S_q \times S_{kv}$ score matrix, which a fused kernel keeps on chip.

Arithmetic intensity is $AI = F/B_{\min}$ for F the FLOP count, and an operator is memory-bound exactly when

$$AI < AI_{\text{ridge}} = \text{FLOP}_{\text{peak}} / \text{BW}_{\text{peak}}, \quad (2)$$

which is 200.6 FLOP/byte for bf16 on this device and 12.5 for fp32, giving Stage 1 a device-derived rather than an arbitrary decision boundary. Attention straddles that boundary: intensity grows with S during prefill but collapses by three orders of magnitude in single-token decode, so the same operator family appears on both sides.

3.5. Baselines

Each operator is measured along four paths with the same timer: eager PyTorch; torch.compile (Inductor), which fuses operations and emits Triton, and is the appropriate baseline for a claim about compiler-generated code; flex_attention, which slots a user-supplied score modification and block mask into a tuned FlashAttention template; and GraphSynth. We also report fused causal scaled_dot_product_attention on identically shaped tensors as a hand-tuned reference, noting it does not implement the semantics under test. flex_attention outputs pass the same gate. Speedups are latency ratios, aggregated as geometric means.

4. Results

4.1. Control experiment: single-tensor operators

Table 1 measures seven single-tensor operators at 8192×8192 in fp32, each moving $B_{\min} = 536.9$ MB against a DRAM time floor of $345 \mu\text{s}$. All 7/7 were synthesized and verified.

Every path converges to the same ceiling: all twenty-one measurements fall in 392 to $398 \mu\text{s}$, i.e. 86.8 to 88.2% of peak, indistinguishable from the 88.6% of a pure device copy. For an operator costing one read and one write this is the hardware limit. The two operators with genuine headroom, eager softmax at 75.4% and layernorm at 58.4%, which materialize intermediates for their reductions, are closed identically by both systems ($1.15\times$ and $1.49\times$); the five pointwise operators were already at the ceiling and correctly report $1.00\times$.

GraphSynth does not outperform Inductor here: 1 of 7 operators, geometric mean $1.00\times$. We report this as a negative result, and as the expected one, since these operators lie inside Inductor’s fusion coverage and both systems hit the same memory wall. Its value is that the methodology reports parity where theory demands parity, which is the precondition for interpreting Table 2.

4.2. Attention variants outside the fused catalog

PyTorch’s fused attention path supports three configurations: unmasked, causal, and grouped-query. Table 2 measures ten variants outside it. All 10/10 produced a numerically verified kernel.

GraphSynth beats eager PyTorch and torch.compile on every operator, by $4.65\times$ to $28.51\times$ (geometric mean $10.83\times$) and $1.41\times$ to $8.91\times$ (geometric mean $3.04\times$) respectively. Inductor is not a weak baseline: it delivers $3.18\times$ to $4.29\times$ over eager on every operator (geometric mean $3.56\times$), fusing the score

Table 1: Single-tensor operators (control). fp32, [8192, 8192]; $B_{\min} = 536.9$ MB; DRAM time floor 345 μ s. A pure device copy reaches 88.6% (389.6 μ s) and is the practical ceiling. “compile” is torch.compile (Inductor). All latencies in μ s, medians of three repetitions.

Operator	Eager PyTorch		compile		×	GraphSynth		GraphSynth vs.	
	T	η_{BW}	T	η_{BW}		T	η_{BW}	eager	compile
relu	393	87.9%	392	88.2%	1.00×	394	87.7%	1.00×	0.99×
leaky_relu	393	87.9%	393	87.7%	1.00×	396	87.2%	0.99×	0.99×
sigmoid	393	87.8%	394	87.7%	1.00×	393	87.8%	1.00×	1.00×
silu	393	87.8%	394	87.7%	1.00×	393	87.9%	1.00×	1.00×
gelu	393	87.9%	395	87.4%	0.99×	392	88.0%	1.00×	1.01×
softmax	458	75.4%	397	87.0%	1.15×	397	86.9%	1.15×	1.00×
layernorm	591	58.4%	396	87.2%	1.49×	398	86.8%	1.49×	0.99×
geometric mean					1.08×			1.08×	1.00×

Table 2: Attention variants outside PyTorch’s fused-kernel catalog. bf16, $B=2$, $H=32$, $S=2048$, $D=128$. Latencies in μ s. “compile” is torch.compile and “flex” is flex.attention; “model” gives the faster of the two LLM backends (2.5=gemini-2.5-pro, 3.1=gemini-3.1-pro-preview). All columns are computed from a single measurement run. Fused causal FlashAttention on identical tensors: 457 μ s. “n/a” marks operators flex cannot express, as they replace softmax normalization rather than modifying scores. Sorted by speedup over eager.

Operator	Baseline latency			GraphSynth		GraphSynth speedup vs.		
	eager	compile	flex	T	model	eager	compile	flex
block-diagonal mask	8251	2580	261	289	3.1	28.51×	8.91×	0.90×
sliding-window mask	8330	2582	226	433	3.1	19.24×	5.96×	0.52×
ReLU attention	8534	1994	n/a	473	2.5	18.05×	4.22×	n/a
ALiBi bias	11076	2623	641	841	3.1	13.16×	3.12×	0.76×
local (banded) mask	8390	2640	402	662	3.1	12.68×	3.99×	0.61×
logit soft-capping	10645	3133	847	901	3.1	11.81×	3.48×	0.94×
sigmoid attention	8534	1991	n/a	1073	2.5	7.95×	1.86×	n/a
prefix-LM mask	8335	2579	557	1390	3.1	6.00×	1.86×	0.40×
per-head temperature	9521	2637	614	1864	2.5	5.11×	1.41×	0.33×
dilated mask	8442	2585	571	1817	3.1	4.65×	1.42×	0.31×
geometric mean						10.83×	3.04×	0.55×

computation and removing intermediate materialization, so the residual gap is not an artifact of an unoptimized comparison.

The gap is structural rather than incidental. Inductor fuses the operations the user wrote but does not change the algorithm: a dense `attn_mask` argument still forces computation of all S^2 scores followed by masking. The synthesized kernels instead compute the mask predicate from tile indices and bound the key/value loop so that fully masked blocks are never visited. The effect is largest where the mask is sparsest: block-diagonal masking (four 512-token documents) reaches 289 μ s and sliding-window masking (width 256, $\approx 12\%$ density) 433 μ s. Both are *faster than FlashAttention itself* (0.63×

and 0.95×

 of the 457 μ s fused causal reference), which is not a contradiction: both masks are strictly sparser than causal, so a block-skipping kernel does strictly less work. flex.attention is faster than GraphSynth wherever it applies, and we report it. On the eight operators it can express it wins all eight (geometric mean ratio 0.55×

, i.e. flex.attention is 1.81×

 faster), narrowly on soft-capping (0.94× and block-diagonal masking (0.90× and widely on dilated masking (0.31×. This is the expected consequence of template reuse: flex.attention inherits an expert-tuned FlashAttention schedule, whereas GraphSynth must rediscover that tuning. flex.attention cannot, however, express two of the ten: its abstraction modifies *scores* but fixes softmax as the normalization, so sigmoid and ReLU atten-

4.3. Sparsity is specified but not executed

tion lie outside it whatever modification is supplied. The template compiler is faster inside its expressible set; the synthesizer has a larger expressible set. Across eleven mask patterns characterized in Stage 1 (sliding-window, dilated, local, block-sparse, block-diagonal, prefix-LM, Longformer, BigBird and others), measured latency varied by under 6% (1707 to 1802 μ s at $S=2048$) while the underlying densities differ by more than 2×

4.4. Synthesis convergence

All eleven dispatched to the same cuDNN masked path, and a sliding-window mask covering $\approx 12\%$ of the score matrix ran 3.27×

 slower than full causal attention. Mechanisms designed to reduce attention from $O(S^2)$ to $O(S \cdot w)$ therefore keep their asymptotic claim in the model definition while executing densely on this stack. Each operator was allowed at most three iterations, with compilation errors, verification failures and measured latency appended to the prompt on each retry. gemini-3.1-pro-preview accepted 10/10, all on the first iteration, and produced the faster kernel on 7 of 10; gemini-2.5-pro accepted 9/10 at a mean of 1.4 iterations, failing to converge on dilated masking within the budget.

One reported failure was a specification defect: sigmoid attention initially failed for both backends with an identical rel-

Table 3: Shape generalization. Kernels were synthesized at $S=2048$ only. “Correct” counts kernels passing the full correctness gate at that S ; geometric means are over that backend’s correct kernels.

Backend	S	Correct	Geo. mean
2.5-pro	1024	5/9	$5.71\times$
	2048	9/9	$6.77\times$
	4096	5/9	$7.42\times$
3.1-pro-preview	1024	10/10	$4.79\times$
	2048	10/10	$6.60\times$
	4096	9/10	$8.85\times$

ative error of ≈ 0.8 , traced to our reference applying the mask before the sigmoid rather than after. An error identical across models and iterations is diagnostic of the specification, not the generated code.

The feedback loop is visible in the released traces. On logit soft-capping, iteration 1 emitted `tl.math.tanh`, absent from this Triton version; iteration 2, given that error, used the removed `tl.dot(trans_b=True)`; iteration 3, given both, compiled and verified. No generated kernel was edited by hand.

4.5. Shape generalization and a silent-correctness hazard

Kernels were synthesized at $S=2048$ only. Re-verifying each accepted kernel at $S \in \{1024, 2048, 4096\}$ (Table 3) shows that for all ten operators at least one backend produced a kernel correct at all three lengths, and that speedups are stable or improve with S . The extreme case is block-diagonal masking at $S=4096$, at $562\ \mu\text{s}$ against $34183\ \mu\text{s}$ eager ($60.88\times$). The advantage is therefore not an artifact of the synthesis shape. The backends differ sharply in shape generality despite the same prompt: four gemini-2.5-pro kernels hardcoded the synthesis shape and raise `AssertionError` at any $S \neq 2048$.

One failure mode deserves emphasis. The gemini-3.1-pro-preview kernel for prefix-LM masking passes the gate at $S=1024$ ($E_{\text{rel}}=1.5 \times 10^{-2}$) and $S=2048$ (1.2×10^{-2}), but at $S=4096$ returns a 23% relative error. It does not raise, does not return the wrong shape, and does not produce NaN: it silently computes the wrong answer. An acceptance gate evaluated at one shape therefore does not certify a kernel at another: a deployed system must re-verify per shape, or restrict each cached kernel to its verified configuration. A second pathological case: one ReLU-attention kernel verifies at all three lengths while running at $\approx 0.20\times$ of eager at every one, so a gate on correctness alone admits performance regressions.

5. Limitations

`flex.attention` is faster where it applies. On the eight of ten operators expressible as a score modification plus block mask it wins all eight (geometric mean $1.81\times$). We make no claim of performance superiority over it; our claim is coverage, 10/10 versus 8/10, and interface, since GraphSynth consumes a natural-language operator description whereas `flex.attention` requires the pattern hand-written in a prescribed form plus a separately constructed block mask. Inside PyTorch’s existing fusion coverage we claim nothing: GraphSynth beats `torch.compile` on 1 of 7 single-tensor operators.

On measurement and scope, everything reported here was re-measured with the single validated timer of Sec. 3.2, superseding the mixed-instrument figures of our earlier report. All results come from one A100-SXM4-40GB, forward pass only, with a single synthesis trial per cell, so we report no variance. Synthesized kernels are also not reliably shape-general, as Sec. 4.5

shows.

6. Related Work

cuDNN and cuBLAS provide hand-tuned kernels for a fixed operator set with no capability for novel topologies. Ansor and AutoTVM search tiling parameters within existing templates but cannot synthesize new kernel logic. MLIR and Triton provide the compiler infrastructure GraphSynth builds on, and PyTorch 2 the graph-compilation substrate. PyTorch’s `flex.attention` is the closest prior system, compiling a user-supplied score modification into a tuned FlashAttention template; our measurements show it is faster than GraphSynth wherever the pattern is expressible in that form. GraphSynth’s novelty is combining synthesis of new kernel logic, which is not bound to a template and therefore covers replaced normalizations that `flex.attention` cannot express, with hardware-measured refinement and numerical verification as a mandatory gate.

7. Conclusion

GraphSynth closes the loop between LLM kernel synthesis and hardware measurement, with numerical verification as a mandatory gate. On ten attention variants outside PyTorch’s fused-kernel catalog it produced verified Triton kernels for all ten, improving on eager PyTorch by a geometric mean of $10.83\times$ and on `torch.compile` by $3.04\times$, with two kernels faster than PyTorch’s hand-tuned FlashAttention path. A single-tensor control experiment reports parity with `torch.compile` at 87 to 88% of peak bandwidth, confirming the methodology does not manufacture speedups where the memory wall forecloses them. Where `flex.attention` applies it is $1.81\times$ faster; GraphSynth’s contribution is the larger expressible set and the absence of any hand-written kernel specification. Finally, a kernel verified at one sequence length can be silently incorrect at another, so per-shape re-verification is a requirement for any system deploying LLM-generated kernels.

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